

Detecting invariant expanding cones for generating word sets to identify chaos in piecewise-linear maps.

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Abstract

We show how the existence of three objects, Ω_{trap} , $\mathbf{W}$, and C , for a continuous piecewise-linear map f on $\mathbb{R}^N$, implies that f has a topological attractor with a positive Lyapunov exponent. First, $\Omega_{\text{trap}} \subset \mathbb{R}^N$ is trapping region for f . Second, $\mathbf{W}$ is a finite set of words that encodes the forward orbits of all points in Ω_{trap} . Finally, $C \subset T\mathbb{R}^N$ is an invariant expanding cone for derivatives of compositions of f formed by the words in $\mathbf{W}$. We develop an algorithm that identifies these objects for two-dimensional homeomorphisms comprised of two affine pieces. The main effort is in the explicit construction of Ω_{trap} and C . Their existence is equated to a set of computable conditions in a general way. This results in a computer-assisted proof of chaos throughout a relatively large regime of parameter space. We also observe how the failure of C to be expanding can coincide with a bifurcation of f . Lyapunov exponents are evaluated using one-sided directional derivatives so that forward orbits that intersect a switching manifold (where f is not differentiable) can be included in the analysis.

1 Introduction

Piecewise-linear maps form canonical representations of nonlinear dynamics and provide effective models of diverse physical systems [10]. Much work has been done to identify properties that imply a piecewise-linear map has a chaotic attractor. These include Markov partitions [15], homoclinic connections [26], and situations where the dynamics is effectively one-dimensional [21]. In the context of ergodic theory, an attractor with an absolutely continuous invariant measure exists for piecewise-expanding maps [7, 35] and piecewise-smooth maps with certain expansion properties [31, 38]. However, when applied to the two-dimensional border-collision normal form (2d BCNF) such properties have only been verified over regions

of parameter space that are small in comparison to where numerical simulations suggest chaotic attractors actually exist [12, 13]. To address this issue we use invariant expanding cones to bound Lyapunov exponents. We show that this appears to be a highly effective method for identifying chaotic attractors in a formal way.

For a continuous map f on $\mathbb{R}^N$, the Lyapunov exponent $\lambda(x, v)$ characterises the asymptotic rate of separation of the forward orbits of arbitrarily close points x and $x + \delta v$:

$$\|f^n(x + \delta v) - f^n(x)\| \sim e^{\lambda_n} \|\delta v\|. \quad (1.1)$$

A positive Lyapunov exponent for bounded orbits is a standard indicator of chaos [24]. Now suppose there exist $N \times N$ matrices A_i such that

$$f^n(x + \delta v) - f^n(x) = \delta A_{n-1} \cdots A_1 A_0 v + o(\delta), \quad (1.2)$$

where $o(\delta)$ vanishes faster than δ as $\delta \rightarrow 0$. These matrices exist if f is C^1 : each A_i is the Jacobian matrix Df evaluated at $f^i(x)$. If $\mathbf{M}$ denotes the set of all A_i and there exists $c > 1$ and a cone $C \subset T\mathbb{R}^N$ with the property that

$$Mv \in C \text{ and } \|Mv\| \geq c\|v\|, \text{ for all } v \in C \text{ and all } M \in \mathbf{M}, \quad (1.3)$$

then immediately we have $\lambda(x, v) > 0$ for any $v \in C$. This idea is attributed to Alekseev [1] (see [5, 37]) and is useful for establishing splitting and hyperbolicity in smooth dynamical systems [8, 27, 34].

Condition (1.3) is rather strong as it requires C to be invariant and expanding for every matrix in the possibly infinite set $\mathbf{M}$. But if f is piecewise-linear then $\mathbf{M}$ contains only as many matrices as pieces in the map. For this reason invariant expanding cones are perfectly suited, and perhaps under-utilised, for analysing piecewise-linear maps. Invariant cones were central to Misiurewicz's strategy for establishing hyperbolicity and transitivity in the Lozi map [26]. More recently in [16] an invariant expanding cone was constructed for the 2d BCNF to finally prove the widely held conjecture that a chaotic attractor exists throughout a physically-important parameter regime $\mathcal{R}_{\text{BYG}}$ that was first highlighted in [4].

In this paper we present a simple but powerful generalisation of the above approach and use it to show that chaotic attractors persist beyond $\mathcal{R}_{\text{BYG}}$, in fact in some places right up to where there exist stable low-period solutions. The idea is to let $\mathbf{M}$ consist of certain products of the A_i , rather than of the A_i themselves. Each product $M \in \mathbf{M}$ is characterised by a word $\mathcal{W}$ connecting its constituent matrices to the pieces of f . As long as the set of all such words $\mathbf{W}$ generates the symbolic itinerary of the forward orbit of x , then again (1.3) implies $\lambda(x, v) > 0$. This approach is quite flexible because there is a large amount of freedom in our choice of the set of words $\mathbf{W}$.

To prove f has a chaotic attractor one needs to identify a trapping region $\Omega_{\text{trap}} \subset \mathbb{R}^N$ (which ensures there exists a topological attractor), a set of words $\mathbf{W}$ that generates the symbolic itineraries for all $x \in \Omega_{\text{trap}}$, and an invariant expanding cone C for the matrices corresponding to the words in $\mathbf{W}$. Below we find these objects for the 2d BCNF. More precisely, we propose a way by which Ω_{trap} and C can be constructed for a particular word set $\mathbf{W}$ and prove that all three objects have the required properties if a certain set of computable